

Sharika P K

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Education

Vellore Institute of Technology

Sep 2023 – 2027

B.Tech (CSE specialization in AI and ML); CGPA: 9.20/10

Skills

Languages: Python, Java, SQL

ML / Deep Learning: TensorFlow, PyTorch, CNN, YOLOv11, GAN, Random Forest

Data, AutoML & Visualization: PyCaret, Data Analytics

Backend & Full-Stack: Spring Boot, React, MySQL, FastAPI

Deployment & Cloud: Streamlit, AWS

Projects

EventOps – Full-Stack Event Management Platform

Apr 2026 – Jun 2026

Full-Stack & Applied ML Project

[GitHub](#)

- Built a full-stack event management platform (Onam, Vishu, college tech fests) with a Spring Boot backend (JWT/RBAC, MySQL with Flyway) and a React frontend.
- Enabled **real-time task and status updates** across teams via WebSockets (STOMP protocol).
- Built a FastAPI ML microservice for **task risk scoring** and z-score based **expense anomaly detection**, within a finance and budgeting module.
- **Tech Stack:** Java, Spring Boot, React, MySQL, FastAPI

AutoML Insurance Cost Prediction

Jan 2026 – Feb 2026

Machine Learning Project

[GitHub](#)

- Engineered an end-to-end AutoML regression pipeline (PyCaret) to predict insurance charges from demographic and health data, automating model comparison and 10-fold cross-validation.
- Selected a GradientBoostingRegressor as the final model, achieving **MAE 2,831**, **RMSE 4,833**, and **R² of 0.83** on cross-validation.
- Visualized actual vs. predicted costs and key risk factors via Power BI; deployed the model as an interactive Streamlit app.
- **Tech Stack:** Python, PyCaret, Scikit-learn, Power BI, Streamlit

Predictive Maintenance for Aircraft Engines

Dec 2025 – Mar 2026

Machine Learning and Deep Learning Project

[GitHub](#)

- Collaborated in a 6-member team to build a predictive maintenance system for aircraft engines (NASA C-MAPSS dataset) for RUL regression and failure classification.
- Built and tuned a CNN regression model (Conv1D, batch norm, dropout) achieving **MAE 11.54**, **RMSE 17.64**, and **R² of 0.60** for RUL estimation.
- Developed a CNN failure-classification model (**98.48%** accuracy); validated on a real-world NGAFID dataset with a tuned Random Forest classifier (**90%** accuracy, **F1 score of 0.88**).
- Built an interactive Streamlit dashboard visualizing engine health, predicted RUL, and maintenance alerts.
- **Tech Stack:** Python, CNN, Random Forest, Streamlit

Certificates and Achievements

- Applied Machine Learning in Python Certificate (Coursera)
- Google IT Support Certificate
- Team Nexus – Advanced to Round 2, Health Hackathon organized by VIT Bhopal & Johns Hopkins University (2025).

Extra Curricular

- **Event Management Lead, Malayalam Club**, VIT Bhopal University
- NSS Volunteer during Higher Secondary (+2)